# Quiz 3 — Sample 2

DATA 412/612

Weeks 1–13. The new material is R Markdown presentations, chunk options, tables, and citations.

Write your answers in an R script or a short `.Rmd`. You may use R and RStudio. You may use your notes and the course website (`https://amousavi1.github.io/612/data-612.html`). You may not use Google, any other website, or an LLM. You do not have to knit.

1. For each goal, name the output format from this course and write the YAML `output` (or Quarto `format`) line you would use.

   - HTML slides you can open in a browser.
   - A PDF slide deck that needs LaTeX.
   - A file someone else will click through in PowerPoint.

   For the HTML choice, say what marks the start of a new slide. If you choose xaringan instead of ioslides, say that slides are separated by a line that contains only `--`.

2. Match each goal to one of `eval = FALSE`, `echo = FALSE`, or `include = FALSE`.

   - The reader must see the code, and R must not run it.
   - R must run the chunk, and the reader must see the plot but not the code.
   - R must load the packages, and the reader must see nothing from that chunk.

3. Write a Quarto plot chunk labeled `fig-hwy` that draws highway mileage against engine size for `mpg`, with the caption `Highway mileage and engine size`. In the next paragraph, refer to that figure with a Quarto cross-reference. Then write the bookdown equivalent: the chunk option `fig.cap` and the `\@ref` call, for a chunk labeled `hwy`.

4. In `mpg`, reorder `class` by median `hwy` and make a boxplot of `hwy` by that factor. Put the plot in a chunk that has a caption, and send a one-row count of the reordered levels to `knitr::kable()` with a caption. The count table should be visible. The plot code should not.